

Formula:

1. Function Point, $FP = \text{count total} \cdot [0.65 + 0.01 \cdot \sum F_i]$

where, $F_i = \text{value adjustment factor } (0 \leq F_i \leq 5)$

count total = sum of all FP entries from complexity table

2. Weighted Methods per class, $WMC = \sum C_i$

where, C_i is the complexity of each method

3. Method Inheritance Factor, $MTF = \frac{\sum M_i(C_i)}{\sum M_a(C_i)}$,

where, $M_i(C_i)$ is the number of methods inherited in class C_i

$M_a(C_i)$ is "total" " " "

4. Coupling Factor, $CF = \frac{\sum_i \sum_j \text{is-client}(C_i, C_j)}{TC^2 - TC}$,

$\text{is-client}(C_i, C_j) = 1$, if there's a relationship between C_i and C_j

$TC = \text{total number of classes}$.

5. Halstead's Software Science Metrics:

program length: $N = n_1 \log_2 n_1 + n_2 \log_2 n_2$

program volume: $V = (N_1 + N_2) \log_2 (n_1 + n_2)$

volume ratio: $L = \frac{2}{n_1} \cdot \frac{n_2}{N_2}$

$n_1 = \# \text{ of distinct operators}$

$N_1 = \text{total operator occurrences}$

$n_2 = \# \text{ of distinct operands}$

$N_2 = \text{total operand occurrences}$

6. Cyclomatic Complexity, $CC = E - N + 2$

$E = \# \text{ of edges}$

$N = \# \text{ of nodes}$

7. Defect Removal Efficiency, DRE = $\frac{E+D}{E}$

E = Errors found before delivery

D: " " after "

8. Feasibility Point Approach:

$$\text{Effort} = \text{Feature Points} \times \text{Productivity rate}$$

9. COCOMO-II model: (Application Composition Model)

$$\text{Effort (person-months)} = (\text{object points}) / \text{productivity rate}$$

10. Early Design & Post-Architecture Model (COCOMO-B)

$$\text{Effort} = A \times (\text{size})^B \times \Pi(EM)$$

A = Tuning coefficient, B: Scale Factor,

$\Pi(EM)$: product of effort multipliers, size: KLOC

11. Software Equation:

$$E = \left[\frac{\text{LOC} \times B^{0.333}}{P} \right]^3 \times \frac{1}{t^4}$$

t = project duration in months

P = " productivity parameter

B = special skills factor, LOC ✓
KLOC x

simplified version:

$$t_{\min} = 8.14 \left(\frac{Loc}{P} \right)^{0.43}, \text{ in months for } t_{\min} \geq 6 \text{ months}$$

$$E = 180 B t^3 \text{ in person-months, for } E \geq 20 \text{ person-months}$$

12. Expected cost = $\sum (\text{path probability}) \times (\text{estimated path cost})$

[Make-Buy Decision]